Engineering AI for Educational Equity

Global Leaders & Scholars in STEM (GLASS)

Final Paper

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March 2026

Abstract

Rapid advances in artificial intelligence and software-driven engineering are reshaping nearly every sector of society, yet access to the benefits of these technologies remains unequal. Education sits at the center of this divide. While digital tools have the potential to personalize learning, expand access to information, and support students across geographic and economic boundaries, they can also reinforce existing inequities when designed without inclusion in mind. Globally, 2.6 billion people still lack internet access, and approximately 60 percent of primary schools worldwide remain unconnected (UNESCO, "UNESCO Spotlights"). This paper looks at how artificial intelligence can be intentionally leveraged to promote global access to quality education. Drawing on professional experiences in software and machine learning, as well as international technology projects, this work explores the gap between technological capability and educational reach. By connecting technical innovation with human-centered design, this paper argues that engineers and computer scientists have both the tools and the responsibility to help close educational gaps. A specific case is made for AI-powered adaptive tutoring platforms, such as Khan Academy's Khanmigo, as models for equitable, scalable educational technology. This analysis aligns with the United Nations Sustainable Development Goal 4: Quality Education and emphasizes that educational equity is not solely a social challenge, but also a systems-level engineering problem.

Introduction

Education has long been viewed as a pathway to opportunity and economic opportunity. In theory, modern technology should make education more accessible than ever before. Digital platforms, intelligent tutoring systems, and online resources promise to remove barriers of

location and cost, offering learning opportunities to anyone with an internet connection. In practice, however, these benefits are unevenly distributed. According to the UNESCO Institute for Statistics, 272 million children worldwide remain out of school, and access to reliable technology, digital literacy, and culturally relevant learning tools varies widely across and within countries (UNESCO Institute for Statistics).

The COVID-19 pandemic laid bare the severity of this divide. UNESCO's Teacher Task Force reported that roughly 826 million students displaced from classrooms during the pandemic lacked access to a household computer, and 706 million had no internet at home (UNESCO, "Startling Digital Divides"). In sub-Saharan Africa, 89 percent of learners lacked household computers and 82 percent had no internet access. These figures show that the digital divide is not a marginal concern, it is a structural barrier that shapes the educational trajectories of hundreds of millions of students.

What makes this moment especially significant is the growing role of artificial intelligence in shaping how people learn. AI-driven systems now recommend content, adapt difficulty levels, automate feedback, and generate explanations. Research indicates that adaptive learning enabled by AI has been associated with improvements in student test performance by as much as 62 percent, and AI usage in general has been linked to a 30 percent enhancement in student outcomes (Zhai and Shi). These tools are powerful, but they are often developed for well-resourced institutions, corporate environments, or users who already possess strong technical foundations. As a result, the same technologies that could democratize education risk widening the gap between those who can fully engage with them and those who cannot.

This paper approaches educational inequality not as a failure of technology, but as a design and deployment problem. By looking at how engineers build intelligent systems in professional and global contexts, this work explores how similar tools could be reimagined to support accessible learning. Rather than treating education as a separate domain from engineering, this paper frames learning environments as complex systems that require thoughtful technical and cultural consideration.

Background and Global Context

The challenge of unequal access to education is explicitly addressed in the United Nations Sustainable Development Goal 4, which calls for inclusive and equitable quality education for all. Despite progress in enrollment and literacy rates, large disparities remain in educational quality, resources, and outcomes. In many regions, students lack access to trained teachers, up-to-date materials, or supportive learning environments. The OECD's 2023 Digital Education Outlook observes that digital inequalities operate across multiple levels: not only in terms of physical access to devices and connectivity, but also in terms of digital skills, usage patterns, and the ability to translate digital engagement into tangible educational outcomes (OECD, "Digital Equity and Inclusion").

The scale of the connectivity gap is substantial. UNESCO Director-General Audrey Azoulay noted in 2025 that 2.6 billion people, representing 32 percent of the global population, still lack internet access, with 1.8 billion of them living in rural areas. In education specifically, approximately 60 percent of primary schools, half of lower secondary schools, and a third of upper secondary schools globally are not connected to the internet (UNESCO, "UNESCO

Spotlights”). These statistics reveal that the infrastructure necessary for digital education is absent in precisely the communities that would benefit most from it.

At the same time, artificial intelligence has become increasingly integrated into everyday tools. Recommendation systems, predictive models, and adaptive algorithms are now commonplace in industry. The global AI-in-education market, valued at approximately $1.82 billion in 2021, is projected to grow at a compound annual rate of 36 percent through 2030 (Zhai and Shi). These technologies excel at personalization and scalability, two features that are especially valuable in education. However, when educational technology is designed without attention to equity, it can reproduce biases.

Understanding this tension requires looking beyond education alone. The ways engineers are trained to optimize systems, measure success, and deploy technology directly influence how educational tools are built. Tawil and Miao advocate for a human-centered approach to steering digital education that counter-balances dominant techno-solutionist thinking. Their framework emphasizes that the use of digital technology should enhance human capacity rather than undermine it, and that connectivity initiatives must be paired with competency development and open-access, inclusive content (Tawil and Miao 51-58). By examining engineering practices across corporate and global projects, this paper highlights how the same technical frameworks can either reinforce inequality or help dismantle it, depending on how they are applied.

Literature Review

Adaptive Learning and AI in Education

A growing body of research demonstrates that AI-powered adaptive learning systems can meaningfully improve educational outcomes when implemented thoughtfully. Ezzaim conducted

a comprehensive literature review documenting the integration of AI and machine learning in e-learning platforms, finding that adaptive systems leverage data-driven algorithms to dynamically adjust content, delivery, and pace of instruction based on individual learner performance and engagement. Their review of 698 papers confirmed that such personalization promotes effective and efficient learning while enhancing student engagement (Ezzaim).

A scoping review by Jumat, published in Heliyon and encompassing 69 studies from 2012 to 2024, found consistent evidence that personalized adaptive learning positively impacts teaching and learning outcomes across higher education contexts. The review highlighted key characteristics including self-paced learning, real-time feedback, and flexibility in content delivery (Jumat). Similarly, a meta-analysis by Wang examining AI-enabled adaptive learning systems from 2010 to 2022 found that these platforms produced meaningful gains in learner outcomes, with personalization emerging as the strongest predictor of improved performance (Wang 451-78).

Intelligent tutoring systems (ITS) represent one of the most extensively studied applications of AI in education. A 2025 systematic review of AI-driven ITS in K-12 education found that personalization and adaptivity are core components influencing effectiveness, and that systems providing real-time, data-driven feedback produced better learning outcomes than traditional instructional methods. The review noted that one large-scale deployment of a cognitive tutor for algebra produced improvements equivalent to roughly an additional year of instruction, though results varied by grade level and implementation context.

The Digital Divide and Educational Inequality

The digital divide in education is well-documented and multidimensional. Van Deursen and Helsper, as cited in the OECD's 2023 working paper on digital equity and inclusion, conceptualize digital divides as operating across three levels: access to technology (first-level), skills and usage patterns (second-level), and the ability to convert digital engagement into tangible outcomes (third-level). Students from lower socioeconomic backgrounds are disproportionately affected across all three levels, with limited access to devices, weaker support networks, and fewer opportunities to develop digital competencies (OECD, "Digital Equity and Inclusion").

The pandemic-era shift to remote learning did not create new inequalities so much as amplify existing ones, making pre-existing digital divides newly consequential for educational outcomes. As the OECD notes, for distance education to be effective, students need a suitable study environment with digital equipment, sufficient digital skills, involved parents, and a well-prepared school, and these circumstances are strongly socially stratified (OECD, "Digital Equity and Inclusion").

The intersection of gender and geography further compounds these challenges. UNESCO has reported that women worldwide are 23 percent less likely to use mobile internet than men, with the gap widening to 30 to 50 percent in parts of Asia, Africa, and South America (IEEE Connecting the Unconnected). In low-income regions, many young people have never touched a computer, and fewer than 40 percent of youth in high-income countries demonstrate even minimal digital literacy proficiency (GVMUN).

Algorithmic Bias and Ethics in Educational AI

The deployment of AI in education raises significant ethical concerns, particularly around algorithmic bias. Baker and Hawn, in a foundational review published in the International Journal of Artificial Intelligence in Education, documented how educational algorithms can produce disparate outcomes across demographic groups. Their review found evidence of bias affecting students based on race, ethnicity, gender, nationality, socioeconomic status, and disability (Baker and Hawn 1052-92). For example, predictive algorithms used to identify at-risk students in higher education have been shown to underestimate the success potential of Black and Hispanic students while overestimating that of White and Asian students (Bird).

The OECD's Digital Education Outlook reported a specific case in which an automated essay scoring algorithm rated native speakers of Arabic, Hindi, and Spanish lower than students from other countries relative to human graders. In response, the test developer implemented a hybrid system combining AI and human evaluation (OECD, "Algorithmic Bias"). This example illustrates both the risks of unchecked algorithmic deployment and the possibility of designing fairer systems through intentional oversight.

Addressing these biases requires a multi-stakeholder approach. The FairAIED framework proposed by researchers emphasizes the integration of fairness metrics throughout the AI development lifecycle, from data collection and algorithm design through deployment and monitoring (FairAIED). The U.S. Department of Education's Office of Educational Technology has similarly called on developers to proactively test AI products in education to mitigate the risk of algorithmic discrimination, though concrete guidance for conducting such tests remains limited (U.S. Department of Education).

Existing Solutions and Organizations in the Field

Several organizations are leading efforts to harness AI for educational equity. Khan Academy, a nonprofit educational platform, developed Khanmigo, an AI-powered tutoring and teaching assistant built on large language models, which has expanded rapidly from pilot programs to deployment across 266 school districts in the United States. In the 2024 to 2025 academic year, usage grew from 40,000 to 700,000 K-12 students, with projections to surpass one million the following year (Khan Academy Rolls Out). Khanmigo employs a Socratic method of guided questioning rather than providing direct answers, and offers free teacher-support tools while charging districts $15 per student per year for student-facing features (Sal Khan Wants).

UNICEF's Giga initiative, in partnership with the International Telecommunication Union, aims to connect every school to the internet by 2030, focusing on least developed countries. The World Bank has committed over $25 billion to education technology projects since 2020, including platforms for hybrid learning and teacher training (GVMUN). UNESCO's Global Education Coalition, launched during the pandemic, brings together over 90 public and private sector partners to develop universal and equitable digital learning solutions (UNESCO, "Startling Digital Divides"). These initiatives represent the ecosystem of actors working to bridge the gap between technological potential and educational access.

Case Study: Engineering Intelligent Systems in Professional Practice

Professional experience in software engineering and machine learning offers valuable insight into how intelligent systems are designed, optimized, and deployed. In corporate environments, AI-driven tools are built to optimize workflows, predict user behavior, and

automate decision-making. These goals emphasize performance metrics like speed, accuracy, and user engagement. While not inherently incompatible with education, they shape how problems are framed and what constitutes success.

Working with data-informed systems reveals their power when paired with clear objectives and well-defined user populations. At the same time, it exposes critical limitations. Systems built for productivity-focused or profit-driven contexts frequently overlook the needs of users with limited resources, different learning styles, or nontraditional educational backgrounds. The assumptions embedded in system design, about connectivity, device access, language proficiency, and prior knowledge, often reflect the demographics of the teams that build them rather than the populations that most need them.

Translating these tools into education requires fundamentally rethinking success metrics. Instead of speed or engagement alone, educational systems must prioritize comprehension, inclusivity, and long-term growth. Research supports this reorientation: Thompson and Miller caution that efficiency-focused AI systems risk sacrificing essential learning objectives such as collaboration, emotional growth, and creativity (Thompson and Miller 156-73). Rodriguez emphasized the need for research exploring the influence of AI tools on critical thinking and higher-order cognitive outcomes, arguing that current evaluations remain too narrowly focused on achievement measures.

This case study suggests that engineers already possess the technical skills needed to support educational equity. The challenge lies in redirecting those skills toward learning-centered outcomes and designing systems that adapt to students, rather than expecting students to adapt to technology.

Global Technology Projects

International technology projects provide a different perspective on education and access. Working in global contexts highlights how learning often occurs outside formal classrooms and how technology can support awareness, problem-solving, and community engagement. Projects focused on health, transportation, or environmental monitoring demonstrate that education can be embedded within systems that people interact with daily.

These experiences reveal the importance of cultural context and local constraints. A solution that works in one region may fail in another due to differences in infrastructure, language, or social norms. UNESCO's Rewired Global Declaration on Connectivity for Education articulates the first principle of inclusive digital education strategies as centering the most marginalized, meaning that digital tools should be designed with underserved populations in mind from the outset rather than as an afterthought (UNESCO, “The Rewired Global Declaration”). For educational technology, this means that AI tools must be flexible and responsive to diverse environments.

Cross-cultural engineering work reinforces the idea that learning tools should empower users to engage with their own realities rather than impose external models of success. In Sudan, for example, the Learning Passport initiative has been used to maintain educational continuity during conflict, and in Nigeria, locally produced digital media has demonstrated the viability of culturally embedded educational content. These examples underscore that effective educational technology must be co-developed with the communities it serves.

Discussion: Designing AI for Educational Equity

Designing AI systems for education requires shifting from a purely technical mindset to a human-centered one. Algorithms must be transparent, accessible, and adaptable. Low-cost hardware, offline functionality, and intuitive interfaces can dramatically expand who benefits from educational technology. Tawil and Miao advocate for integrated digital education systems that prioritize teacher support, address connectivity issues through competency strengthening, and provide open-access, inclusive quality digital learning content (Tawil and Miao 51-58).

Equally important is involving educators and learners in the design process, ensuring that tools align with real needs rather than abstract assumptions. Khan Academy's development of Khanmigo offers a compelling model here. The platform was designed in partnership with educators, incorporating teacher feedback into its lesson-planning, rubric-generation, and student-monitoring features. Its Socratic tutoring approach, guiding students to discover answers rather than providing them directly, represents a pedagogically grounded alternative to the tendency of AI tools to simply deliver information. The fact that Khanmigo is being piloted in countries such as India, Brazil, and the Philippines suggests the potential for cross-cultural applicability, though challenges of language support, connectivity, and computational cost remain significant (Khan Academy Rolls Out).

Ethical considerations are central to this discussion. Data privacy, algorithmic bias, and unequal access to training data all shape educational outcomes. Baker and Hawn note that unknown biases represent a particularly insidious challenge: in some cases, we do not know that bias exists for a specific group, and in others, we know bias is present but do not understand how it manifests (Baker and Hawn 1052-92). Engineers must recognize that neutrality in design is often an illusion and that intentional choices are necessary to promote fairness. When AI systems are built with equity as a core objective, incorporating diverse training data, regular bias audits,

and fairness-aware evaluation metrics, they can support personalized learning without reinforcing existing hierarchies.

One specific, actionable proposal emerging from this analysis is the development of lightweight, open-source adaptive tutoring platforms designed explicitly for low-resource environments. Inspired by Khan Academy's model but engineered for offline functionality, multilingual support, and minimal hardware requirements, such a platform could bring personalized AI tutoring to communities currently excluded from the benefits of educational technology. Microsoft's collaboration with Khan Academy to explore small language models based on Phi-3, which can perform well for simpler AI tasks and are more cost-effective than large models, points toward the technical feasibility of this approach (Khan Academy and Microsoft). If such models can operate locally on devices without constant cloud connectivity, the barriers to deploying adaptive learning in rural and underserved areas would be substantially reduced.

Reflection and Personal Insight

Reflecting on experiences across professional, academic, and global settings highlights a recurring theme: technology is most powerful when it helps people connect to knowledge and to each other. Exposure to diverse learning environments reveals how education is shaped by systems far beyond the classroom, including infrastructure, policy, and design choices.

Participation in GLASS has reinforced the idea that global challenges require interdisciplinary thinking. Educational inequality is not solely a pedagogical issue, but a systems problem that engineers are uniquely positioned to address. As a computer science student with professional experience in software and machine learning, I have seen firsthand how the

technical skills used to build recommendation engines and predictive models could be redirected toward adaptive tutoring systems, accessibility tools, and offline learning platforms. The same neural network architectures that power content recommendation in commercial applications can be adapted to recommend learning pathways tailored to individual student needs, prior knowledge, and cultural context.

The example of Khanmigo is particularly resonant. It demonstrates that AI tutoring need not be a luxury reserved for affluent school districts. With thoughtful engineering, including the use of smaller, more efficient language models, open educational content, and community-driven design processes, it is possible to imagine a future in which every student, regardless of geography or socioeconomic status, has access to a patient, adaptive, always-available tutor. This is not a plug-and-play implementation; it requires sustained collaboration between engineers, educators, policymakers, and local communities. But the technical foundations exist, and the moral imperative is clear.

Recognizing this responsibility reshapes how engineering success is defined, moving beyond technical achievement toward social impact. The question is not whether engineers can build intelligent educational tools; they already can. The question is whether they will choose to build them for everyone.

Future Directions

This paper has argued that artificial intelligence and engineering can play a meaningful role in closing global education gaps, but only when designed with intention and care. Technological capability alone is insufficient. Equity must be treated as a core design constraint,

not an afterthought, embedded in every stage of the development process from data collection to deployment and evaluation.

The evidence reviewed in this paper supports several key conclusions. First, AI-powered adaptive learning systems have demonstrated consistent positive effects on student outcomes across diverse educational contexts, with personalization and real-time feedback emerging as the most impactful features. Second, the digital divide remains a structural barrier of enormous scale, affecting billions of people and disproportionately impacting students in low-income, rural, and marginalized communities. Third, algorithmic bias in educational technology is not hypothetical; it has been documented across multiple demographic categories and application areas, but it can be mitigated through intentional design, diverse training data, and ongoing auditing.

Future work should prioritize three areas. First, partnerships between engineers, educators, and communities must be strengthened to co-create learning technologies that are accessible, culturally aware, and sustainable. The success of Khan Academy's district partnerships and UNICEF's Giga initiative demonstrates that cross-sector collaboration is essential to scaling educational technology equitably. Second, research and development should focus on lightweight AI models capable of operating in low-connectivity, low-resource environments. The exploration of small language models like Microsoft's Phi-3 family represents a promising direction that could make adaptive tutoring accessible to populations currently excluded from such tools. Third, regulatory frameworks and ethical guidelines must evolve to keep pace with AI deployment in education, ensuring that student data is protected, algorithmic decisions are transparent, and the benefits of AI are distributed equitably.

As AI continues to shape how knowledge is produced and shared, engineers have an opportunity, and an obligation, to ensure that these systems expand opportunity rather than restrict it. At its core, educational equity is both a technical and a human challenge, one that demands thoughtful, responsible engineering. The tools exist. The research supports their potential. What remains is the collective will to deploy them in service of those who need them most.

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